



Miro1 Protein in Fibroblasts from Parkinson's Patients

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Project 129

Summary

Miro1 is an outer mitochondrial membrane protein that anchors mitochondria to microtubule motors and is removed from damaged mitochondria to stop mitochondrial motility as an early step prior to the clearance of dysfunctional mitochondria. Our previous findings discover an accumulation of Miro1 on damaged mitochondria as a convergent phenotype in fibroblasts from a small cohort of familial and sporadic PD patients (Hsieh et al., 2016). In this study, we examine this Miro1 phenotype in the first-released case-control PD cohort from PPMI in a blind fashion.

Method

Cell culture

Fibroblasts were cultured in high-glucose DMEM (SH30243.01, Invitrogen) supplemented with 10% fetal bovine serum (F0926, Sigma-Aldrich) and maintained in a 37°C, 5% CO₂ incubator with humidified atmosphere. The media were refreshed every 3-4 days, and split every 7-8 days.

CCCP treatment

CCCP (C2759, Sigma-Aldrich) was prepared at 40 mM in DMSO fresh every time and applied at 40 μM in fresh culture medium (1:1000) for 6 hrs.

Mitochondrial Isolation

CCCP/DMSO treated fibroblasts were lifted by a cell scraper, and mechanically homogenized with a Dounce homogenizer in 750 μl isolation buffer (200 mM sucrose, 10 mM TRIS/MOPS, pH 7.4). After centrifugation at 500 g for 10 min, crude supernatant was spun at 10,000 g for 10 min to pellet intact mitochondria. Mitochondrial pellet was washed twice with isolation buffer. After this step, supernatant was referred to “cytosolic fraction”, and pellet was resuspended in 50 μl lysis buffer (50 mM Tris, pH 8.0, 150 mM NaCl, and 1% Triton X-100–T8787, Sigma-Aldrich) with 0.25 mM phenylmethanesulfonylfluoride (P7626, Sigma-Aldrich) and protease inhibitors (Roche) referred to “mitochondrial fraction”.

SDS-PAGE and Western blotting

Fibroblast mitochondrial lysates were mixed 1:1 with 2X laemmli buffer (4% SDS, 20% Glycerol, 120 mM Tris-HCl, 0.02% bromophenol blue, 700 mM 2-mercaptoethanol) and boiled for 5 min prior to being loaded into an SDS-PAGE. 10% polyacrylamide gels (acrylamide:bis-





acrylamide=29:1) and Tris-Glycine-SDS Buffer (24.8 mM Tris, 192 mM glycine, 0.1% SDS) were used for electrophoresis. After electrophoresis, nitrocellulose membranes (1620115, Bio-Rad) were used in wet transfer with Tris-Glycine buffer (24.8 mM Tris, 192 mM glycine) at 360 mA for 2 hrs on ice. Transferred membranes were first blocked overnight in phosphate-buffered saline (PBS) containing 5% fat-free milk and 0.1% tween-20 at 4°C, and then incubated with the following primary antibodies: mouse anti-Miro1 (WH0055288M1, Sigma-Aldrich) at 1:1,000, rabbit anti-Miro1 (HPA010687, Sigma-Aldrich), rabbit anti-VDAC (4661S, Cell Signaling Technology) at 1:1,000, at 4°C overnight in blocking buffer. West Dura ECL Reagents (34075, GE Healthcare) were used for ECL immunoblotting with HRP-conjugated goat anti-mouse or rabbit IgG (Jackson ImmunoResearch Laboratories) at 1:5-10,000. Membranes were exposed to UltraCruz autoradiography films (Santa Cruz Biotechnology) and developed on a Konica Minolta SRX-101A developer.

Statistical Analysis

Digital images were scanned from films in a 16-bit tiff format. The intensity of protein bands was measured by ImageJ (ver. 1.48V, NIH). The intensity of Miro1 band is normalized to that of the mitochondrial loading control protein VDAC from the same blot, and expressed as a fraction of the mean of our universal healthy control line with DMSO treatment (line 1815 in Hsieh et al., 2016); this control was included in every independent experiment. N=4 independent experiments. The band intensities of VDAC are not significantly different among all fibroblast lines and conditions ($p = 0.9871$). The distribution of data points is expressed as (1) mean and (2) standard error of the mean. The Mann-Whitney U test was performed for comparing normalized Miro1 intensities within the same subject (DMSO v.s. CCCP), and the one-way ANOVA test was performed for VDAC intensities. Statistical analyses were performed using Prism software (ver. 5.01, GraphPad).

References

Hsieh CH, Shaltouki A, Gonzalez A, Betterncourt da Cruz A, Burbulla L, St. Lawrence E, Schule B, Krainc D, Palmer T, Wang X. Functional impairment in Miro degradation and mitophagy is a shared feature in Familial and Sporadic Parkinson's Disease. *Cell Stem Cell*. 2016. <http://dx.doi.org/10.1016/j.stem.2016.08.002>

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